Limits of Resolution in Small Machine-Learning Benchmarks: A Noise Budget with a Quantum Case Study

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October 2026

Abstract

Quantum machine learning (QML) benchmarks on small tabular datasets are usually read as rankings of architectures, often from differences of a few hundredths of AUC. We ask how large a difference such a benchmark can resolve and how much the verdict depends on the inferential procedure. On a Western Mediterranean precipitation panel (four stations, 1938–2026; 528 test station-months), one AUC has a standard error of about 0.024, 4–7 times the seed standard deviation of Random Forest, XGBoost and an LSTM, so more seeds cannot remove the uncertainty of a finite test set. The minimum detectable difference of 15-seed means is 0.031–0.044 AUC and falls only as $N^{-1/2}$ with test size. Near this limit the verdict depends on the procedure: for Random Forest–LSTM (+0.015) the year-block bootstrap interval excludes zero while DeLong does not ($p = 0.19$), and known-truth simulations show that each procedure can be miscalibrated under plausible dependence between test years. In a case study with matched fitting rows and 15 seeds per model, four hybrid quantum models scored 0.020–0.026 AUC below Random Forest and were not distinguished from a parameter-matched, circuit-free classical control; deciding whether the circuit adds anything would need five to seven times more test data. No model, classical or quantum, improved on a month×station climatology, and within stations the climatology discriminated best. We propose a reporting protocol for small QML benchmarks: a stated estimand, a minimum detectable difference computed before ranking models, separate seed and sampling error, a base-rate reference and a classical control at the same interface.

Keywords: Quantum Machine Learning, Benchmarking, Statistical Power, Minimum Detectable Effect, Seed Variance, Precipitation Forecasting.

1 Introduction

Claims of quantum utility on classical tabular data frequently rest on a single training run per model, capacity-asymmetric classical baselines, and uncertainty estimates that cover only one source of error (Bowles et al., 2024). The machine-learning literature has long pointed out that a single run confounds initialization luck with architectural merit (Henderson et al., 2018; Bouthillier et al., 2021), and that tests computed on one fixed test set ignore the variance induced by the training procedure (Dietterich, 1998; Demšar, 2006). Variational circuits aggravate the issue: in the noisy intermediate-scale quantum (NISQ) era (Preskill, 2018), parameterized quantum circuits are sensitive to initialization and, at larger qubit counts, prone to barren plateaus (McClean et al., 2018), and they are usually combined with a classical backbone that compresses the input into a few qubits (Huang et al., 2021; Schuld & Killoran, 2022).

Throughout, "noise" means statistical variability (test-set sampling error and optimization-seed variance), not quantum hardware noise. This paper asks a design question that comes before any ranking: how large a difference in AUC can a benchmark with a few hundred test cases resolve, and how much does the verdict on a small difference depend on the inferential procedure? The answer is largely a property of the evaluation design rather than of the application. The standard error of one AUC is set by the size and prevalence of the test set (about 0.024 for the 528 test cases studied here); it is several times the variation between training seeds of the classical models, and it falls only as $N^{-1/2}$, so repeating training with more seeds does not remove it. For differences of a few hundredths, the paired standard error, the dependence between test cases and the choice of procedure decide the verdict. Variational hybrids add a second complication: their seed variance is large enough that seed and sampling error must both be reported.

We work through these questions on a deliberately modest but realistic case: a multi-decadal, multi-station Western Mediterranean precipitation panel, an area where heavy precipitation is linked to cut-off lows aloft (DANAs; Romero et al., 1999; Nieto et al., 2005) and where tree ensembles are strong baselines on engineered tabular features (Grinsztajn et al., 2022). The task is a 1-month-ahead probabilistic forecast of precipitation exceeding a pooled regional 60th percentile (P_{60} = 28.1 mm) from antecedent 12-month features. The panel has the features that make small benchmarks hard to read: a few hundred correlated test cases, strong base-rate structure across sites and seasons, and labels affected by missing data. It is a testbed for evaluation methodology, not an operational early-warning system.

Applied climate- and weather-QML studies have addressed separate parts of the problem: opportunities for climate modeling (Schwabe et al., 2025a), parameter-count and initialization-matched comparisons for cloud cover (Schwabe et al., 2025b; Pastori et al., 2026), wind-speed forecasting (da Silva et al., 2026), quantum kernels for climate emulation (Bazgir & Zhang, 2024), synthetic time-series benchmarks (Fellner et al., 2026), variational data assimilation (Kumar et al., 2026), and parameter-matched neural comparisons (Marchisio et al., 2026). In the time-series and weather QML studies cited above, quantum models are reported as better than, or as having the potential to outperform, classical baselines on the basis of point estimates: Marchisio et al. (2026) report a 15.4 % lower MSE than a parameter-matched classical leaky-integrate-and-fire neuron on a 2,000-sample test set, with one value per model and no interval or test, and da Silva et al. (2026) use 10 runs per configuration and show their spread in violin plots, but compare the models through single error values with no interval or test, although their test sets span only 14 and 5 days. We do not claim that these effects are spurious; the point is that the evidence reported cannot separate them from sampling and seed noise. This paper complements these studies by asking how

large a difference an evaluation design can resolve.

What the paper offers to other benchmarks. The statistical tools are standard (DeLong, cluster bootstrap, minimum detectable difference); the contribution is to combine them around two explicit estimands and to quantify what they imply for benchmark design. Three results transfer beyond this panel. First, a resolution rule, checked empirically on the panel: the minimum detectable AUC difference (MDE) of two models compared on the same N test cases falls as $N^{-1/2}$, from about 0.03 for 15-seed means at $N \approx 500$ to about 0.016 at $N \approx 2{,}000$ (Section 5.4). Second, near this limit verdicts depend on the procedure, and known-truth simulations show under which dependence structures each procedure is miscalibrated (Section 4.2). Third, a reporting protocol whose items need only the test predictions of the models (Table 13).

What the paper does not claim. It does not rank the architectures. In particular, it does not show that Random Forest, or tree ensembles in general, beat quantum or neural models on tabular data: the Random Forest–hybrid gap is resolved by some procedures and not by others, Random Forest only matches a month×station climatology, and within a station the climatology discriminates best (Section 5.2). For the same reason the panel is a poor testbed for choosing an architecture, since on it no learned model adds resolvable skill; it is used here for the statistical question, for which small differences between models are the case of interest (Section 5.3).

Contributions.

1. **A noise budget with two explicit estimands**, the difference of single-run AUCs and of seed-averaged AUCs, with paired DeLong, year-block bootstrap and two-way (years and seeds) bootstrap inference, for three classical models and four hybrid quantum models on 528 test station-months (Sections 3 and 4.1).
2. **Known-truth simulations** of two correlated scorers with shared and model-specific year effects, which check the type-I error, coverage and power of the procedures at this sample size (Section 4.2).
3. **A resolution analysis for benchmark design**, combining a closed-form approximation with an empirical curve of the minimum detectable difference against test size (Section 5.4).
4. **A QML case study and a reporting protocol**: four hybrid quantum models, a circuit-free classical control at the same four-dimensional interface, a standalone LSTM and Random Forest, fitted on the same rows with 15 seeds each, compared against climatological references on discrimination, proper scoring and within-station discrimination, with an audit of label validity (Sections 4.3 to 4.5 and 5.5).

Further material (full label audit, positive control, single-seed analyses, historical runs with other architectures, ablations, operational metrics, circuit costs) is in the Supplementary Material.

2 Data, Task and Models

2.1 Dataset, Target and Split

The study domain is the Spanish Mediterranean coast (Catalonia, the Valencian Community and the Region of Murcia), a transition zone between Mediterranean (*Csa*) and semi-arid (*BSk*/*BSh*) Köppen

regimes subject to severe convective episodes (Romero et al., 1999; Nieto et al., 2005). Daily records of four AEMET stations, Alacant/Alicante, Barcelona Aeropuerto, València and San Javier Aeropuerto (Table 1), were obtained from *datosclima.es* under the AEMET Open Data framework (downloaded on 1 June 2026; station metadata from the AEMET climatological normals pages, accessed 28 October 2026). They are aggregated to monthly values per station: precipitation sum, wettest day, days above the daily 90th percentile, mean station-level pressure (not reduced to sea level) and its variability, temperature and diurnal range, wind speed and maximum gust, sunshine duration, and reference evapotranspiration from FAO-56 Penman–Monteith with humidity estimated from the minimum temperature, radiation from sunshine duration and a constant wind speed $u_2 = 2\,\mathrm{m\,s^{-1}}$ (Allen et al., 1998). Months with fewer than 10 valid daily records are excluded; monthly totals are sums over the available days, so incomplete months remain in the panel (Section 4.5). The panel has 3,363 station-months; after a 12-month antecedent window per station it yields 3,315 prediction windows. Inhomogeneities from relocations or instrument changes remain an accepted limitation (Alexandersson, 1986; Peterson et al., 1998).

Each window uses months $i, \ldots, i+11$ (35 features: 12 precipitation lags, 17 engineered physical features, 2 cyclic month terms and 4 station indicators) to predict whether precipitation in month $i+12$ exceeds the pooled regional 60th percentile, $P_{60} = 28.1$ mm, computed on the training partition. The split is chronological: training 1938–1996 (2,273 windows), validation 1997–2007 (514) and test 2008–2026 (528; 41.1 % positive). Because records are joined across five daily variable files, San Javier and Valencia leave the panel when their pressure series end (2010 and 2012), and 83.7 % of the test windows come from Alicante and Barcelona. Test base rates differ between stations (29–56 %), so part of the pooled AUC reflects telling stations apart (Section 4.4). All scaling and anomaly statistics use pre-1997 data only; computing the anomalies this way rather than from the full record moves the AUC of each model by 0.003 or less (Section S2.4). The test period falls in a time of accelerated warming, so the evaluation takes place under a covariate shift whose effect cannot be isolated with one chronological split.

Table 1: AEMET stations (climatological identifier, altitude), station-month windows by partition (training 1938–1996, validation 1997–2007, test 2008–2026) and test-set class prevalence for the regional exceedance threshold ($P_{60} = 28.1$ mm). Coordinates: Alicante 38°22′21″ N 0°29′39″ W; Barcelona 41°17′30″ N 2°3′38″ E; San Javier 37°47′20″ N 0°48′12″ W; Valencia 39°28′50″ N 0°21′59″ W. Daily records span 1938-09 to 2026-05 (Alicante), 1961-01 to 2026-05 (Barcelona), 1951-01 to 2010-12 (San Javier) and 1938-02 to 2012-02 (Valencia).

Station	AEMET id (alt.)	Train	Val	Test	Total	Test positive rate
Alicante	8025 (81 m)	671	132	221	1,024	29.4% (65 / 221)
Barcelona	0076 (1 m)	378	132	221	731	50.2% (111 / 221)
San Javier	7031 (4 m)	535	132	36	703	36.1% (13 / 36)
Valencia	8416 (13 m)	689	118	50	857	56.0% (28 / 50)
Total		**2,273**	**514**	**528**	**3,315**	**41.1% (217 / 528)**

2.2 Models and Fitting Sets

Models in the main analyses. Two references, a month×station climatology fitted on the training partition and persistence (lag-1 precipitation), are compared with three classical models, Random Forest (300 trees, depth 8; Breiman, 2001), XGBoost (500 trees, depth 4) and a standalone bidirectional LSTM (10,103 parameters), and with four hybrid quantum–classical models simulated without noise in PennyLane.

LSTM+QNN compresses a 51-dimensional vector (the 16-dimensional LSTM state concatenated with the 35 features) to 4 rotation angles that feed a 4-qubit circuit, which computes a truncated Fourier series of these angles (Schuld et al., 2021); it has 2,613 parameters, 24 of them quantum. LSTM+PI-QNN is the same model trained with an additional hand-set penalty loosely motivated by the Clausius–Clapeyron relation. Its constants are not derived from physics and are not dimensionally consistent, and in the implementation the penalty acts as a fixed calendar prior against confident rain forecasts in winter and spring (Section S4). We therefore treat LSTM+PI-QNN as a regularization variant of LSTM+QNN, not as a physics-informed model, and draw no conclusion about physics-informed learning from it. LbL-QNN a 3-layer circuit trained layer by layer (Skolik et al., 2021), and LSTM+QTL a quantum head on a frozen pre-trained LSTM (Mari et al., 2020). The classical control has the architecture of LSTM+QNN with the circuit replaced by a linear $4 \to 4$ map (20 parameters), so it isolates the contribution of the circuit at the same interface. The models differ in capacity: the standalone LSTM has 10,103 trainable parameters, the hybrids 2,613–2,625 (LSTM+QTL trains 617 on top of its frozen LSTM) and the control 2,609. Comparisons of a hybrid with the standalone LSTM or with Random Forest are therefore comparisons of whole pipelines and cannot be attributed to the circuit; the comparison that isolates the circuit is the one with the control, whose parameter count matches that of LSTM+QNN to within four parameters. Architectures and hyperparameters are given in Sections S2.7 and S3, and parameter counts in Table S10. Logistic regression, gradient boosting, a quantum-kernel SVM (whose product-state kernel factorizes classically) and further bottleneck and encoding variants appear only in the historical runs of the Supplement (Section S7).

Fitting sets and seeds. All models in the main analyses use the same 2,273 training rows, the neural and hybrid models use the 514 validation rows for early stopping, and all are scored on the same 528 test rows, with 15 seeds each; the seed index is nominal across frameworks. The Random Forest and LSTM fits of the classical and quantum experiments are identical (their test probabilities agree to within 10^{-11}). The primary metric is the ROC-AUC, complemented by the Brier score (Brier, 1950). Historical runs in the canonical pipeline, which standardized anomalies with the full record and fitted the tree ensembles on training plus validation (2,787 rows), are reported only descriptively; the extra rows change the Random Forest AUC by 0.003–0.007 (Section S2.4).

3 Statistical Framework

3.1 Estimands and Inference

For models A and B with test probabilities $p_{A,s}$, $p_{B,s}$ from seeds $s = 1, \ldots, S$ ($S = 15$), the main test statistic of this paper is the difference between seed-averaged test AUCs,

$$\hat{\Delta} = \frac{1}{S}\sum_s \widehat{\mathrm{AUC}}(p_{A,s}) - \frac{1}{S}\sum_s \widehat{\mathrm{AUC}}(p_{B,s}), \tag{1}$$

and, for single-run comparisons, the difference of two individual AUCs. Sampling inference targets the corresponding performance difference for the evaluation population, conditional on the observed training data and on the fitted seed-specific models. The DeLong variance additionally conditions on the observed test years and treats station-months as exchangeable; the year-block bootstrap and the simulations of Section 4.2 instead target the difference for new years, so a DeLong miscalibration in the simulations partly reflects this change of target. A bootstrap that also resamples seeds targets the seed-averaged difference

Table 2: Map of experiments: which pipeline, fitting rows, seeds and stored probabilities stand behind each result, and its inferential status.

Experiment	Pipeline	Fitting rows	Seeds	Stored test probabilities	Use (tables)
Core QML	train-only	2,273 (all models)	15	all seeds	primary (Tables 6 to 8)
Classical rerun	train-only	2,273 (RF and LSTM as in the core)	15	all seeds	primary (Tables 3 and 4)
Post hoc re-analyses	saved predictions	none (no retraining)	15	all seeds	secondary (Tables 9, 10 and 12)
Label sensitivity	saved predictions	none (no retraining)	15	all seeds	sensitivity (Tables S15 and S17)
Historical runs	canonical	trees 2,787; neural 2,273	15 (AUCs only)	seed 0 and single run (seed 42)	descriptive (Section S7)

over the distribution of optimization seeds, again conditional on the training data. The mean of AUCs is linear in the DeLong structural components, so the paired sampling variance of $\hat{\Delta}$ is obtained by averaging the components over seeds; this treats the seeds as fixed.

3.2 Sampling Uncertainty

Test-set sampling error is estimated in three ways. (i) The paired DeLong estimator for correlated ROC curves (DeLong et al., 1988; Hanley & McNeil, 1982) treats station-months as independent and may therefore be optimistic for monthly data. (ii) A calendar-year block bootstrap resamples the 19 test years (2008–2026) with all stations of a year together ($B = 2{,}000$; Künsch, 1989; Obuchowski, 1997). Bootstrap confidence intervals are the 2.5th and 97.5th percentiles of the replicated statistic, and bootstrap standard errors are the sample standard deviations of the replicates; resamples that contain only one class are discarded. Where a bootstrap p-value is reported, it is $2\min\{\Pr(\hat{\Delta}^* \le 0), \Pr(\hat{\Delta}^* \ge 0)\}$ over the replicates $\hat{\Delta}^*$, bounded below by $1/B$. (iii) A two-way bootstrap resamples years and, within each model, seeds. With only 19 clusters the block bootstrap is itself imprecise: cluster-based variance estimates are biased downward when clusters are few (Cameron et al., 2008), the simulations of Section 4.2 show a liberal block bootstrap (type-I error 0.074–0.084 at nominal 0.05), and for some pairs it returns a smaller standard error than DeLong. A rejection by the cluster procedures alone is therefore treated as provisional. All procedures are reported, and none is assumed a priori to be the correct one for the observational panel. As a check on the block standard error, a delete-one-year jackknife recomputes $\hat{\Delta}$ with each of the $K = 19$ test years removed, $\mathrm{SE}_{\mathrm{jk}} = \{(K-1)/K \sum_k (\hat{\Delta}_{(k)} - \bar{\Delta}_{(\cdot)})^2\}^{1/2}$, with intervals $\hat{\Delta} \pm t_{0.975,18}\,\mathrm{SE}_{\mathrm{jk}}$ (Section 5.6).

3.3 Seed Variability and the Seed Share

For a model with seed standard deviation σ_{seed} and mean sampling standard error SE of one AUC we report $\sigma^2_{\mathrm{seed}}/(\sigma^2_{\mathrm{seed}} + \mathrm{SE}^2)$. This is a descriptive ratio of two conditional variances (seeds, on a fixed test set; test sampling, for a fixed trained model), not a decomposition of the total uncertainty of a training procedure; it compares the scale of the two sources. For a difference between models A and B, whose seeds are drawn independently, the seed variance is $\sigma^2_A + \sigma^2_B$ for one seed per model and $(\sigma^2_A + \sigma^2_B)/S$ for

the difference of S-seed means, and SE is the corresponding paired sampling standard error (for one seed per model, averaged over the $S \times S$ seed pairings).

3.4 Minimum Detectable Difference

For a paired difference with standard error SE_Δ, the minimum detectable difference at two-sided $\alpha = 0.05$ and power 0.80 is

$$\mathrm{MDE} = (z_{0.975} + z_{0.80})\,\mathrm{SE}_\Delta \approx 2.80\,\mathrm{SE}_\Delta. \tag{2}$$

Under a normal approximation with a locally constant standard error, the MDE describes the sensitivity of the design to a true difference: true differences below it are expected to be declared significant in fewer than 80 % of repeated experiments. It is not a threshold that an observed difference must exceed, and an observed gap between $1.96\,\mathrm{SE}_\Delta$ and $2.80\,\mathrm{SE}_\Delta$ can be nominally significant. We therefore report observed gaps with their intervals and use the MDE only to state which true differences the design would detect reliably.

3.5 Scope of Claims and Multiplicity

The primary analyses are the noise budget, the paired comparisons, the simulations and the comparison with the climatology; none was pre-registered. Holm correction is applied to the DeLong p-values within two families of three seed-mean comparisons (the classical models, and `LSTM+QNN` against Random Forest, the control and the LSTM), and table captions quote the first-step threshold 0.017 for reference. Analyses added after the main analysis are secondary: the comparisons of the other three hybrids (Table 8), the Brier scores (Table 9), the within-station AUC (Table 10), the resolution curve (Table 12) and the block-length sensitivity. Bootstrap intervals are pointwise, and every nominal p-value outside the two Holm families is hypothesis-generating. Seed-level tests in the Supplement (Wilcoxon, Friedman) compare seeds on one fixed test set and describe optimization variance, not generalization.

4 Results

4.1 The Noise Budget

We first measure the two sources of error. On the scale of one AUC, conditional test-sampling variability exceeds across-seed variability for all three classical models (Table 3): one AUC has a standard error of 0.023–0.027 under either estimator, whereas the seed standard deviation is 0.004–0.006, so the seed accounts for 2–7 % of the combined seed-plus-sampling variance (a descriptive comparison of scales, not an estimate of the share of total uncertainty that more seeds could remove). The hybrid quantum models have larger seed standard deviations. The 15-seed `LSTM+QNN` of Section 4.3 has a seed standard deviation of 0.0098 and a seed share of 15 % (12 % with the block standard error; last column of Table 3); for `LbL-QNN`, `LSTM+QTL` and `LSTM+PI-QNN`, fitted in the same experiment, the seed standard deviations are 0.0129, 0.0147 and 0.0170 and the seed shares 23, 27 and 34 % (block 18, 22 and 30 %). On the scale of one AUC the sampling term is the larger one for all the models analyzed, but for the hybrids the seed term is no longer small.

Table 4 and Figure 1 compare the 15-seed means ($\Delta = +0.015$ for RF − LSTM; the Random Forest and

Table 3: Noise budget for one AUC, train-only pipeline, 15 seeds per model, 528 test rows. iid: DeLong; block: calendar-year block bootstrap (B = 2,000, 19 years). The seed share uses the DeLong standard error; with the block standard error it is 2, 4, 5 and 12 %. The classical columns are the rerun of Table 4; `LSTM+QNN` is the core experiment of Section 4.3; all models are fitted on the 2,273 training rows. Pairwise differences in Table 4 are computed from unrounded seed means.

Quantity	Random Forest	XGBoost	Standalone LSTM	LSTM+QNN
Mean AUC (15 seeds)	0.683	0.662	0.668	0.663
SE of one AUC, iid / block	0.023 / 0.026	0.024 / 0.025	0.024 / 0.027	0.024 / 0.027
SD across seeds	0.0035	0.0054	0.0063	0.0098
Seed share of combined variance	**2 %**	**5 %**	**7 %**	**15 %**

LSTM fits are the same as in the core experiment of Section 4.3). The standard error of a paired difference is about half that of one AUC, and averaging the seeds lowers it further, but the DeLong MDE remains 0.031–0.044 (block estimates are 0.034, 0.017 and 0.043). For RF − XGBoost the difference is +0.021 (z = 1.93, p = 0.054) and both bootstrap intervals include zero: weak, borderline evidence. For RF − LSTM both bootstrap intervals exclude zero by a small margin, but DeLong does not (p = 0.189), and the block standard error (0.006) is about half of the DeLong one (0.011). With 19 clusters the block estimate is itself imprecise, and its MDE (0.017) is about half the iid one; given the limited number of clusters and the uncertainty about the dependence structure, neither estimate is treated as definitive. We therefore describe this comparison as dependent on the inferential procedure and not as established or refuted. XGBoost against the LSTM shows no evidence of a difference, and none of the three DeLong comparisons rejects the null after Holm correction over the three pairs (smallest p = 0.054 against a first-step threshold of 0.017). The bootstrap intervals in Table 4 are pointwise 95 % intervals, are not multiplicity-adjusted, and no bootstrap p-values were computed for these pairs.

Pairing removes most of the sampling error shared by two models but none of the seed variance, so the seed share is larger for a difference than for one AUC: for the classical pairs it is 17–22 % of a single-seed difference (block 17–35 %) and 2–3 % (2–9 %) of a difference of 15-seed means. For the hybrids the seed is not a minor term: in the differences of `LSTM+QNN` with Random Forest, the linear control and the standalone LSTM it is 32–47 % for one seed per model (block 45–47 %) and still 6–27 % (block 13–33 %) for the 15-seed means, and 8–47 % (block 16–53 %) for the 15-seed means of the other three hybrids. The DeLong and year-block intervals condition on the fitted seeds and omit this component; the years×seeds bootstrap includes it, which is why its intervals in Table 7 are wider than the year-block ones.

Table 4: Paired comparison of the 15-seed means (train-only pipeline): $\Delta = \overline{\mathrm{AUC}}_A - \overline{\mathrm{AUC}}_B$ with the paired DeLong standard error on seed-averaged components, two-sided p-value, 95 % intervals from the calendar-year block bootstrap (19 years, B = 2,000) and from a two-way bootstrap that resamples years and seeds, and the MDE (2.80× SE) under DeLong and the block bootstrap.

Pair	Δ	SE (DeLong)	z (p)	Block 95 % CI (SE)	Two-way 95 % CI	MDE iid / block
RF − XGBoost	+0.021	0.0110	1.93 (0.054)	[−0.002, +0.045] (0.0122)	[−0.003, +0.046]	0.031 / 0.034
RF − LSTM	+0.015	0.0112	1.31 (0.189)	[+0.002, +0.025] (0.0061)	[+0.001, +0.027]	0.031 / 0.017
XGBoost − LSTM	−0.007	0.0159	−0.42 (0.677)	[−0.039, +0.022] (0.0153)	[−0.037, +0.023]	0.044 / 0.043

Single-seed comparisons are less stable. The paired DeLong standard error of two single runs is 0.013–0.018 (MDE 0.036–0.052), and in the train-only pipeline RF > XGBoost is declared significant in 27 % of the 15 × 15 seed pairings (21 % under the block bootstrap), RF > LSTM in 3 % (24 %) and XGBoost

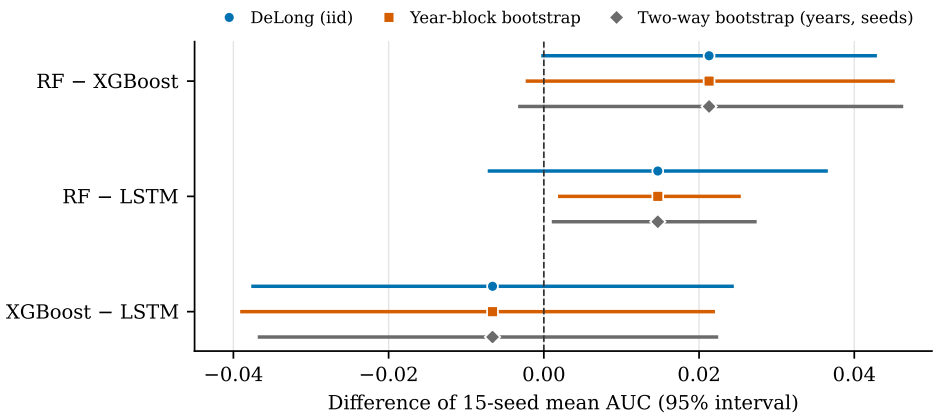


Figure 1: Differences of 15-seed mean AUC with 95 % intervals from the three procedures (train-only pipeline, 528 test rows). The procedures agree for RF − XGBoost and XGBoost − LSTM and disagree on whether the RF − LSTM interval excludes zero.

versus LSTM in 0 % (Table S1). A verdict about a single pair of runs therefore depends largely on which seeds happen to be compared (Section S1.1).

4.2 Simulation with Known Truth

Because DeLong and the year-block bootstrap disagree on gaps near the MDE, we check both on data whose true difference is known. We simulate two scorers whose population AUC difference is known by construction or approximated on large reference samples (four independent draws of 400,000 rows; Monte Carlo standard error of the population difference at most 0.0002). Labels have prevalence 0.41 with year-level effects on the logit, and the 528 rows are split among 19 years as equally as possible or, in one scenario, unequally (multinomial with Dirichlet(3) shares). The scorers are binormal with noise correlation $\rho = 0.9$, which reproduces the paired DeLong standard error of the classical comparisons (0.011), and their design AUCs without year effects are 0.65 and $0.65 + \Delta$. Year effects enter as a prevalence shift (τ_π), a shift shared by both scorers (τ_s) and, in scenarios C–E, a shift specific to each scorer and independent between them (τ_m). Scenario B has $\tau_\pi = \tau_s = 0.3$, C adds $\tau_m = 0.3$, D is C with unequal year sizes, and E has all three equal to 0.6. The target is the pooled population difference for new years drawn with equal weight (details in Section A). When the design difference is 0 the scorers are exchangeable and the population difference is exactly 0. Each scenario has 500 replications; the block bootstrap uses $B = 300$ (Section A). Rejection rates have a Monte Carlo standard error of about 0.01 near 0.05.

Table 5: Simulation with known truth ($N = 528$, 19 years, prevalence 0.41, $\rho = 0.9$, 500 replications). τ_π, τ_s and τ_m: standard deviations of the year effects on the prevalence logit, on a shift shared by both scorers and on a shift specific to each scorer. Type-I error and coverage are for an exactly null difference (DeLong, block bootstrap); SD($\hat{\Delta}$) is the Monte Carlo standard deviation of the estimate, to be compared with the mean standard error (DeLong / block). Last column: power at the largest simulated difference (population difference in brackets). All rows are in the Supplement.

Scenario	Type-I DeLong	Type-I block	Cover. DeLong	Cover. block	SD($\hat{\Delta}$)	Mean SE	Power (DeLong / block)
B: shared year effects ($\tau_\pi = \tau_s = 0.3$)	0.040	0.074	0.960	0.926	0.0108	0.0111 / 0.0108	0.74 / 0.78 (0.029)
C: + scorer-specific year effects ($\tau_m = 0.3$)	0.068	0.076	0.932	0.924	0.0154	0.0144 / 0.0153	0.50 / 0.47 (0.028)
D: C with unequal year sizes	0.096	0.084	0.904	0.916	0.0168	0.0144 / 0.0155	0.51 / 0.48 (0.028)
E: all $\tau = 0.6$	0.220	0.074	0.780	0.926	0.0290	0.0185 / 0.0273	0.35 / 0.21 (0.023)

Table 5 and Figure 2 give three results. First, with year effects shared by the two scorers (scenario B)

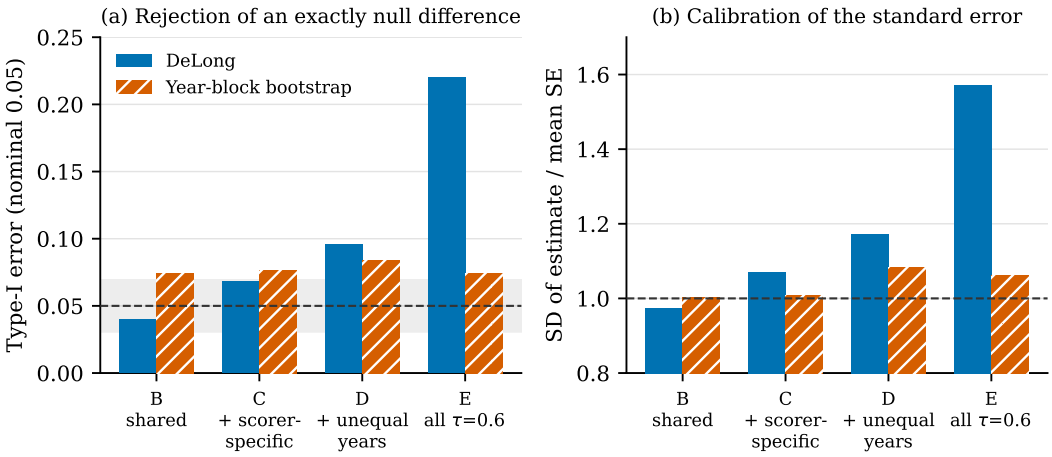


Figure 2: Simulation scenarios B–E (Table 5). (a) Rejection rate of an exactly null difference; the band is the 95 % Monte Carlo interval around the nominal 0.05 for 500 replications. (b) Ratio of the Monte Carlo standard deviation of the estimate to the mean standard error; values above 1 mean that the standard error is too small.

DeLong is calibrated (type-I error 0.040, coverage 0.960), whereas the block bootstrap rejects the exact null in 7.4 % of replications (coverage 0.926). Second, with scorer-specific year effects the DeLong standard error underestimates the standard deviation of the estimate (ratio 1.07 in C, 1.17 in D, 1.57 in E) and its type-I error rises to 0.068, 0.096 and 0.220; the block standard error tracks the standard deviation (ratios 1.01–1.08) and its type-I error stays at 0.074–0.084. Neither procedure is uniformly better calibrated: DeLong fails when year effects are scorer-specific, and the block bootstrap with 19 clusters is less sensitive to the scenario but liberal. Third, power follows the MDE rule where DeLong is calibrated: in scenario B (MDE 0.031) a true difference of 0.029 is detected in 74 % of replications (block 78 %) and 0.0145 in 32 % (35 %). The mechanism is Gaussian and binormal with no spatial structure or optimization noise. A version calibrated to the observed year sizes, prevalence and score correlation of the panel gives DeLong type-I errors of 0.034–0.056 and block errors of 0.054–0.092, an MDE of 0.033, and only 14–22 % power at the observed ensemble gap of 0.0115 (Section S6.1). A positive control that degrades the observed Random Forest probabilities by a known amount reaches 80 % detection at an imposed reduction of about 0.04 (Section S1.3); it perturbs observed probabilities and does not establish coverage or power over new samples.

4.3 Case Study: A Hybrid Quantum Model under the Noise Budget

With these limits established, we ask what the benchmark can say about hybrid quantum models. The core experiment fits Random Forest, the standalone LSTM, the circuit-free control, `LSTM+QNN` and the three further hybrids on the same 2,273 training rows, with 15 seeds each and every test probability stored (Table 6). Sharing rows does not imply equal capacity, tuning budget or input representation; the configuration of each model is given in Table S3. In particular, the standalone LSTM has about four times the trainable parameters of the hybrids, so its advantage over them says nothing about the circuit; the capacity-matched comparison is with the Linear-4D control. Random Forest has the highest mean; every hybrid lies below it and below the standalone LSTM, with seed standard deviations 1.5–2.7 times that of the LSTM. Tables 7 and 8 compare the seed means with the procedures of Section 4.1.

Table 6: 15-seed test AUC (train-only pipeline, same 2,273 fitting rows; mean and sample SD), sorted by mean. The last column is the difference to `LSTM+QNN`; it is descriptive, not a test. The frozen extractor of `LSTM+QTL` is the standalone LSTM of the same seed (identical test probabilities). Climatology: AUC 0.6796.

Model	**Mean AUC**	**SD**	**Gap to LSTM+QNN**
Random Forest (2,273 rows)	0.6829	0.0035	+0.020
Standalone LSTM	0.6682	0.0063	+0.005
LSTM+QNN (24 quantum parameters)	0.6634	0.0098	0
LbL-QNN	0.6632	0.0129	−0.000
LSTM+PI-QNN	0.6601	0.0170	−0.003
LSTM+QTL	0.6565	0.0147	−0.007
Linear-4D classical control	0.6539	0.0078	−0.009

Table 7: Paired comparisons of the 15-seed means with `LSTM+QNN` (train-only pipeline, 528 test rows, 19 test years; same procedures and notation as Table 4). DeLong p-values are unadjusted; the Holm first-step threshold for these three pairs is 0.017. The MDEs are specific to each pair.

Difference	$\hat{\Delta}$	SE (DeLong)	z (p)	Year-block 95 % CI (SE)	Years×seeds 95 % CI	MDE DeLong / block
RF − LSTM+QNN	+0.020	0.0109	1.79 (0.074)	[+0.007, +0.031] (0.0062)	[+0.006, +0.034]	0.031 / 0.017
Linear-4D − LSTM+QNN	−0.009	0.0077	−1.22 (0.223)	[−0.026, +0.006] (0.0081)	[−0.030, +0.008]	0.022 / 0.023
LSTM − LSTM+QNN	+0.005	0.0050	0.97 (0.334)	[−0.003, +0.013] (0.0042)	[−0.007, +0.018]	0.014 / 0.012

Table 8: Paired comparisons of the 15-seed means with the three further hybrids (same experiment, procedures and notation as Table 7). For `LSTM+QTL` the comparator is its own frozen extractor, which is the standalone LSTM of the same seed. Secondary analysis: DeLong p-values are unadjusted and the intervals pointwise.

Difference	$\hat{\Delta}$	SE (DeLong)	z (p)	Year-block 95 % CI (SE)	Years×seeds 95 % CI	MDE DeLong / block
RF − LSTM+PI-QNN	+0.023	0.0116	1.96 (0.050)	[+0.010, +0.036] (0.0066)	[+0.005, +0.039]	0.033 / 0.019
RF − LbL-QNN	+0.020	0.0116	1.70 (0.089)	[+0.004, +0.035] (0.0080)	[+0.002, +0.038]	0.033 / 0.022
RF − LSTM+QTL	+0.026	0.0118	2.25 (0.025)	[+0.012, +0.039] (0.0070)	[+0.010, +0.043]	0.033 / 0.020
Linear-4D − LSTM+PI-QNN	−0.006	0.0069	−0.88 (0.376)	[−0.023, +0.008] (0.0082)	[−0.026, +0.013]	0.019 / 0.023
Linear-4D − LbL-QNN	−0.009	0.0065	−1.41 (0.158)	[−0.027, +0.005] (0.0084)	[−0.031, +0.009]	0.018 / 0.024
Linear-4D − LSTM+QTL	−0.003	0.0075	−0.34 (0.736)	[−0.018, +0.012] (0.0075)	[−0.020, +0.015]	0.021 / 0.021
LSTM − LSTM+PI-QNN	+0.008	0.0055	1.48 (0.139)	[−0.0005, +0.017] (0.0044)	[−0.005, +0.023]	0.015 / 0.012
LSTM − LbL-QNN	+0.005	0.0049	1.04 (0.299)	[−0.008, +0.018] (0.0067)	[−0.011, +0.021]	0.014 / 0.019
Extractor (= LSTM) − LSTM+QTL	+0.012	0.0044	2.69 (0.007)	[+0.003, +0.020] (0.0044)	[−0.002, +0.025]	0.012 / 0.012

Random Forest exceeds the hybrid by 0.020 AUC. The procedures disagree, as for RF − LSTM in Section 4.1: both bootstrap intervals exclude zero (year-block $[+0.007, +0.031]$; years×seeds $[+0.006, +0.034]$), whereas DeLong does not reject the null ($p = 0.074$; Holm-adjusted $p = 0.22$ across the three comparisons of this family), and the year-block standard error (0.0062) is again about half of the DeLong one (0.0109). The bootstrap intervals are pointwise and not multiplicity-adjusted. Under the alternative label definitions of Section 4.5 (Table S17) the difference stays between +0.011 and +0.020, with DeLong p from 0.074 to 0.367.

In the capacity-matched comparison (2,613 against 2,609 parameters) the hybrid is not distinguished from the circuit-free classical control: the control scores 0.009 below the hybrid, and the year-block and years×seeds intervals run from −0.026 to +0.006 and from −0.030 to +0.008, so the data are compatible with a small advantage of either model; no equivalence margin was specified in advance, and this is not an equivalence test. Against the standalone LSTM the difference is +0.005 with intervals that include zero. A bottleneck-free LSTM with the same hidden size, two further 4-dimensional controllers with other activations and full-feature XGBoost, evaluated in the canonical pipeline (Section S7.8), score 0.6435–0.6575. These experiments do not establish the four-dimensional bottleneck as the cause of the observed Random Forest–hybrid gap. The quantum parameters are 0.9 % of the trainable parameters of `LSTM+QNN`, so most trainable parameters are classical, although this ratio alone does not quantify the functional contribution of the circuit.

The three further hybrids behave like `LSTM+QNN` (Table 8). Random Forest exceeds them by 0.020–0.026, with year-block and years×seeds intervals that exclude zero in all three cases and DeLong p of 0.025–0.089; none is distinguished from the Linear-4D control (differences −0.003 to −0.009, all intervals include zero), and `LSTM+PI-QNN` and `LbL-QNN` are not distinguished from the standalone LSTM. `LSTM+QTL` trains a quantum head on a frozen pre-trained extractor, and in each seed the extractor is the standalone LSTM of that seed. The hybrid scores 0.012 below its own extractor (DeLong $p = 0.007$, year-block interval $[+0.003, +0.020]$); the years×seeds interval, which also resamples seeds, includes zero ($[-0.002, +0.025]$). In this experiment, then, replacing the classical head with the quantum head lowered the AUC rather than raising it. With RF − `LSTM+QTL` ($p = 0.025$) and RF − `LSTM+PI-QNN` ($p = 0.0497$), it is one of the three hybrid comparisons in which DeLong and the year-block interval both reject zero at the nominal level; these are secondary analyses and were not adjusted for multiplicity.

4.4 Reference Baselines: Climatology and Probabilistic Skill

The comparisons so far are between learned models. A stricter question is whether any of them beats a forecast that uses only the station and the calendar month. A month×station climatology fitted on the training partition reaches AUC 0.680 and a Brier score of 0.2224, 8.2 % better than a constant forecast at the training base rate (0.2422); persistence is weaker (lag-1 precipitation, AUC 0.589). Random Forest exceeds the climatology by +0.003 AUC (range −0.002 to +0.011 across seeds) and no seed is significantly better under either test; XGBoost and the LSTM score below the climatology on point estimates (−0.018 and −0.011), also without significance. Two further references require no retraining. A station-only base rate fitted on the training partition reaches AUC 0.599 (Brier score 0.2392), about half of the climatology's excess over 0.5; adding the month raises the AUC by 0.080 (DeLong $p < 0.001$). Fitted on the training and validation years (1938–2007), the same climatology reaches AUC 0.686 (Brier score 0.2186), and

the 15-seed Random Forest mean differs from it by −0.003 (DeLong $p = 0.83$; year-block 95 % CI [−0.025, +0.016]): the advantage of Random Forest over the training-only climatology is smaller than what the climatology itself gains from eleven more years (+0.006, $p = 0.27$). On proper scoring (Table 9; 15-seed train-only predictions, no retraining), no model improves on the climatological Brier score on average. Random Forest matches it (mean Brier score 0.2238 against 0.2224; Brier skill score −0.006, from −0.013 to +0.001 across seeds, with one seed of 15 marginally below the climatology; year-block 95 % interval of the difference [−0.006, +0.009]). XGBoost, the standalone LSTM, the Linear-4D control and the four variational hybrids, all trained with positive-class weighting, score 0.017–0.023 worse than the climatology (skill −0.075 to −0.104; every year-block interval excludes zero) and have higher expected calibration errors (ECE 0.120–0.137, against 0.075 for Random Forest; whether they are over- or under-confident was not analyzed). The canonical single-run evaluation, including logistic regression, gradient boosting and the quantum-kernel SVM, gives the same picture (Sections S7.2 and S7.3).

Table 9: Probabilistic skill against the month×station climatology (train-only pipeline, 15 seeds per model, 528 test rows; archived predictions, no retraining; script `exp21_brier_trainonly.py`). Brier score: 15-seed mean (SD). $\mathrm{BSS}_{\mathrm{clim}} = 1 - \mathrm{BS}/0.2224$: seed mean [minimum, maximum]. ΔBS: mean Brier score minus that of the climatology, with year-block 95 % interval ($B = 2{,}000$). ECE: 10-bin expected calibration error, seed mean. The canonical single-run version of this table, which also includes logistic regression, gradient boosting and the quantum-kernel SVM, is Table S20.

Model	AUC	Brier score	$\mathrm{BSS}_{\mathrm{clim}}$	ΔBS [year-block CI]	ECE
Month×Station Climatology	0.680	0.2224	0	–	0.089
Random Forest	0.683	0.2238 (0.0009)	−0.006 [−0.013, +0.001]	+0.001 [−0.006, +0.009]	0.075
XGBoost	0.662	0.2456 (0.0025)	−0.104 [−0.127, −0.085]	+0.023 [+0.009, +0.038]	0.125
Standalone LSTM	0.668	0.2395 (0.0046)	−0.077 [−0.109, −0.033]	+0.017 [+0.007, +0.027]	0.137
Linear-4D control	0.654	0.2429 (0.0042)	−0.092 [−0.117, −0.051]	+0.021 [+0.010, +0.032]	0.131
LSTM+QNN	0.663	0.2421 (0.0061)	−0.088 [−0.126, −0.053]	+0.020 [+0.011, +0.028]	0.131
LbL-QNN	0.663	0.2393 (0.0044)	−0.076 [−0.111, −0.052]	+0.017 [+0.008, +0.025]	0.128
LSTM+PI-QNN	0.660	0.2391 (0.0067)	−0.075 [−0.119, −0.025]	+0.017 [+0.007, +0.026]	0.125
LSTM+QTL	0.657	0.2400 (0.0069)	−0.079 [−0.128, −0.030]	+0.018 [+0.008, +0.027]	0.120

Within-station discrimination. Because the threshold is pooled across stations whose base rates differ, the pooled AUC partly rewards telling stations apart. The within-station AUC counts only positive–negative pairs of months from the same station (the per-station AUCs averaged with weights $n_{\mathrm{pos}} n_{\mathrm{neg}}$); a station-only predictor scores exactly 0.5 on it (Table 10; `exp23_within_station_auc.py`). On this scale every learned model loses 0.020–0.035, the climatology only 0.010, and the month×station climatology has the highest value (0.670). Random Forest is not distinguished from it (−0.010, year-block 95 % CI [−0.034, +0.009]), whereas `LSTM+QNN` is below it (−0.031, [−0.060, −0.005]). The Random Forest − `LSTM+QNN` difference keeps its size (+0.022) but its year-block interval now reaches zero ([−0.0003, +0.039]). Within a station, therefore, none of the learned models adds discrimination beyond the seasonal cycle encoded by the climatology.

4.5 Label Validity under Incomplete Months

Finally, we check whether the conclusions depend on how the labels are defined. Monthly precipitation is the sum of the available daily values, and months with at least 10 valid days are retained, so an incomplete month can only under-count precipitation and some months labeled negative may in fact exceed the

Table 10: Pooled and within-station AUC (train-only pipeline, 15-seed means, 528 test rows; archived predictions, no retraining). Within-station AUC: positive–negative pairs of the same station only. SD: across seeds. Per-station AUC in the last four columns (test n: 221, 221, 36, 50).

Model	Pooled AUC	Within-station AUC (SD)	Alicante	Barcelona	San Javier	Valencia
Month×Station Climatology	0.680	0.670 (–)	0.681	0.663	0.555	0.692
Random Forest	0.683	0.660 (0.005)	0.703	0.628	0.555	0.646
XGBoost	0.662	0.627 (0.007)	0.675	0.586	0.558	0.667
Standalone LSTM	0.668	0.648 (0.012)	0.674	0.628	0.585	0.661
Linear-4D control	0.654	0.620 (0.014)	0.644	0.598	0.619	0.652
LSTM+QNN	0.663	0.639 (0.015)	0.681	0.604	0.590	0.642
LbL-QNN	0.663	0.638 (0.018)	0.677	0.606	0.570	0.646
LSTM+PI-QNN	0.660	0.634 (0.026)	0.674	0.601	0.567	0.659
LSTM+QTL	0.657	0.631 (0.025)	0.657	0.611	0.565	0.654

threshold. Incomplete months are 23.3 % of the test partition (123 of 528; 7.5 % in training), and the inner join across the five daily files also discards observed rain in 121 station-months, mostly in Barcelona. We therefore re-evaluated the saved 15-seed predictions, without retraining, under eight label scenarios (S0–S7): subsets of rows with the project labels, labels from a proportional-coverage sum, and labels from the unjoined precipitation sum (full audit and tables in Section S6.2). The sign of every paired difference is preserved in all scenarios, but not its nominal significance. RF − XGBoost ranges from +0.014 to +0.026 (DeLong p 0.019–0.285), RF − LSTM from +0.005 to +0.015 (p 0.189–0.622; its year-block interval excludes zero only under the original labels, and narrowly), and RF − `LSTM+QNN` from +0.011 to +0.020 (p 0.074–0.367; year-block interval excluding zero in five of the eight scenarios). The differences of the hybrid with the circuit-free control and with the standalone LSTM have intervals that include zero in every scenario.

5 Discussion

5.1 What a Benchmark of This Size Can and Cannot Conclude

Under the iid DeLong approximation the 80 %-power MDE for the classical seed-mean comparisons on 528 correlated test months is 0.031–0.044 AUC, and the observed gaps between the architectures studied (0.003–0.03) are mostly below it. This does not mean that the models are equal. Three observations follow. First, on the scale of one AUC, test-sampling variability exceeds seed variability for the classical models (seeds are 2–7 % of the combined variance) and remains the larger term for the hybrids, but there the seed share rises to 15–34 %, so seed and sampling error must both be reported for variational models. Second, near the MDE the verdict depends on the estimand and on the procedure: single-seed verdicts depend on which seeds are compared, and for the seed means DeLong and the cluster procedures disagree on RF − LSTM and on RF − `LSTM+QNN`. Third, the reference points are strong: Random Forest matches the climatology in AUC and Brier score, the other models are worse on proper scoring, and within a station no model discriminates better than the month×station climatology.

For the quantum results, the four hybrids lie 0.020–0.026 AUC below Random Forest, are not distinguished from a circuit-free classical control at the same interface, and `LSTM+QTL` scores 0.012 below its own frozen classical extractor. The evidence does not establish a quantum performance advantage in this case study; it neither shows nor rules out one on larger or richer data.

5.2 What These Results Do Not Show

Because Random Forest has the highest mean AUC, the results could be read as one more instance of tree ensembles outperforming other models on tabular data (Grinsztajn et al., 2022). They do not support that reading. The gap between Random Forest and `LSTM+QNN` is resolved by the cluster procedures but not by DeLong, and it does not survive multiplicity correction; the gaps to the other three hybrids are secondary comparisons that were not adjusted for multiplicity; among the classical models, Random Forest against XGBoost is borderline and Random Forest against the LSTM depends on the procedure. Random Forest does not improve on a month×station climatology in AUC or in Brier score, and once differences between stations are removed it discriminates no better than that climatology (Table 10). What the data support is that none of the learned models, classical or quantum, adds resolvable skill beyond the seasonal and geographic base rate of this panel, and that the differences between them are below what 528 test months can resolve.

Nor do the results show that variational circuits are ineffective in general: the study concerns 4-qubit circuits behind a 4-dimensional classical bottleneck, simulated without noise, on a small panel whose predictable signal is largely seasonal. And they do not show that DeLong or the block bootstrap is wrong: each is calibrated under some of the simulated mechanisms and not under others, and the panel does not identify which mechanism holds.

5.3 A Low-Signal Testbed

The strongest empirical result of the case study is negative for every model: no learned model improves on the month×station climatology in AUC or Brier score, and once differences between stations are removed all of them lose discrimination while the climatology loses little (Tables 9 and 10). At a one-month lead, with monthly aggregates from four stations, the signal available beyond the seasonal and geographic base rate is therefore close to zero. Several factors may contribute: the limited predictability of monthly precipitation one month ahead from local antecedent data, monthly sums that do not resolve the convective episodes that produce much of the heavy precipitation in this region, and incomplete-month labels. The label scenarios of Section 4.5 preserve the signs of the differences, so the conclusions are not driven by the last factor, but the panel does not allow the three to be separated.

This makes the panel a poor testbed for deciding which architecture is better: on these data the answer is none of them, and a ranking of the learned models would reflect differences of a few thousandths to hundredths of AUC around a common ceiling set by the climatology. For the question of this paper it is a representative case rather than a weakness. Small QML benchmarks typically compare models whose differences are of this size, and the noise budget, the disagreement between procedures and the resolution rule describe what such a benchmark can conclude whatever the reason for the small differences. The statistical argument does not depend on the panel having little signal; the conclusions about particular architectures do, and they should be tested on data where learned models clearly beat a base-rate reference.

5.4 Implications for Benchmark Design

The resolution limits above follow from the size and composition of the test set, not from the climate application, so they can be estimated before a benchmark is run.

For a binary classifier with AUC A, n_1 positives and n_0 negatives, the Hanley–McNeil approximation (Hanley & McNeil, 1982) gives the standard error of one AUC from (A, n_1, n_0) alone; for the present test set it returns 0.024, matching the DeLong value. Assuming a correlation of 0.75 between the AUCs of two models scored on the same rows (the paired standard errors measured here imply 0.72–0.83), Table 11 gives the approximate 80 %-power MDE for other test-set sizes: about 0.049 for $N = 500$ at prevalence 0.41 and about 0.086 at prevalence 0.10. These values are specific to the assumed AUC (0.68), inter-model correlation and iid approximation; an MDE is a design sensitivity and not a bound on detectable effects, and temporal and spatial dependence can change it in either direction for paired comparisons (for RF − LSTM the block standard error here is smaller than the DeLong one). They do not replace validation on further datasets.

An empirical version of this scaling is obtained from the panel itself (Table 12; `exp24_resolution_curve.py`, no retraining): random subsets of 5 to 16 of the 19 test years (300 per size, all stations of a year together) are re-evaluated with the archived predictions. For every pair and every procedure the standard error falls approximately as $N^{-1/2}$ (fitted exponents 0.46–0.53), and the single-run MDEs lie about 10–20 % below the Hanley–McNeil values, so Table 11 is slightly conservative here. Extrapolating with this scaling, the MDE of the 15-seed means would equal the observed Random Forest − `LSTM+QNN` gap (0.020) at about 1,300 test rows under DeLong, 2.5 times the present test set, but already at about 420 rows under the year jackknife, so the two procedures also disagree on how much data the comparison needs. The comparisons that bear on a quantum contribution are much further from resolution: the hybrid against the circuit-free control (−0.009) would need about 2,800–3,500 rows and the hybrid against the standalone LSTM (+0.005) about 3,300–4,400.

Table 11: Approximate MDE for the AUC difference of two models on the same test set (Hanley–McNeil standard error, AUC 0.68, inter-model correlation 0.75, $\alpha = 0.05$, power 0.80). Test-set size N and positive-class prevalence π.

N	MDE, $\pi = 0.41$	MDE, $\pi = 0.10$
250	0.069	0.122
500	0.049	0.086
1,000	0.035	0.061
2,500	0.022	0.038
5,000	0.015	0.027

Table 12: Empirical resolution curve: median 80 %-power MDE over random subsets of test years (300 subsets per size; archived train-only predictions). Single run: paired DeLong MDE of seed s against seed s, averaged over the 15 seeds. 15-seed means: paired DeLong on seed-averaged components and delete-one-year jackknife. The last row is the full test set; HM: Hanley–McNeil single-run value of Table 11 at the same N.

			Single run		**RF − LSTM+QNN, 15-seed means**	
Years	N	**HM**	**RF − LSTM**	**RF − LSTM+QNN**	**DeLong**	**Jackknife**
5	144	0.091	0.077	0.084	0.060	0.035
8	218	0.074	0.060	0.067	0.047	0.027
10	282	0.065	0.053	0.059	0.042	0.024
13	360	0.058	0.046	0.052	0.037	0.021
16	444	0.052	0.042	0.047	0.033	0.019
19	528	0.048	0.038	0.043	0.031	0.017

In practice, the MDE of a planned comparison can be computed from the test size, the prevalence and an assumed AUC before any model is trained (Table 11) and checked afterwards on the archived predictions (Table 12). If the differences of interest are below it, more seeds do not help: averaging seeds reduces the seed component but leaves the sampling error unchanged, and a larger test set is the only remedy. For the comparison that bears on a quantum contribution here, a hybrid against a circuit-free control at the same interface, resolving a difference of 0.01 AUC would need several thousand test cases.

5.5 Recommended Reporting Protocol

Table 13 lists the practices that follow from the results. Items 1, 2 and 4 need only the test probabilities of the models; item 3 needs a specified and executed generating mechanism.

Table 13: Reporting protocol for small QML benchmarks.

#	Practice
1	State the estimand (single run or seed average; conditional on which sources) and report paired confidence intervals for it, with DeLong and a block bootstrap over the natural cluster (here calendar years); report discrepancies between procedures instead of selecting one.
2	Report the MDE as a measure of design sensitivity next to the intervals; do not treat it as a significance threshold or classify gaps below it as uninformative.
3	Check the procedures against a simulation with known differences at the sample size and dependence structure of the study.
4	Report across-seed variability separately from test-sampling uncertainty; if a relative-scale ratio is used, state its conditional and descriptive interpretation, and treat seed-level tests as statements about optimization, not generalization.
5	Include a climatological (or base-rate) reference with a proper scoring rule, and report skill relative to it.
6	Fit all model families on the same rows and tune them under a comparable budget, or state and quantify the asymmetry.
7	Add a classical control at the same low-dimensional interface as the quantum circuit and with a matched parameter count; attribute differences to the circuit only through this control, not through comparisons with larger classical models.
8	Audit label validity (here, incomplete months) and re-evaluate the comparisons under alternative labels.
9	Choose equivalence margins by scientific relevance, independently of the MDE and of the results, include sampling error, and treat an interval wider than the margin as inconclusive.
10	When the target threshold is pooled across heterogeneous sites, report a site-only baseline and a within-site AUC, so that skill is not credited for separating sites with different base rates.

5.6 Limitations

Inferential procedures. The block bootstrap relies on 19 calendar years, and for RF − LSTM and RF − `LSTM+QNN` its standard error is about half the DeLong one; a delete-one-year jackknife agrees with it (0.0062 for both comparisons), and blocks of two to four years change only the RF − XGBoost verdict. With so few clusters, cluster-based variance estimates are biased downward (Cameron et al., 2008), consistent with the liberal block bootstrap in the simulations, and a block standard error below the iid value can arise from negative within-year correlation of the paired-difference components or from the downward bias of cluster-bootstrap variance estimates; it is not evidence of variance reduction, and the simulations do not establish which procedure is calibrated for the panel (Section S6.3). Small-sample corrections for clustered inference, such as the wild cluster bootstrap (Cameron et al., 2008), were not applied; a test period with more independent years, or several independent test periods, would be needed to settle the RF − LSTM and RF − `LSTM+QNN` verdicts.

Hybrid models. The comparisons concern 4-qubit circuits behind a 4-dimensional classical bottleneck in noiseless simulation; the hybrids have about a quarter of the trainable parameters of the standalone LSTM, so only the comparison with the Linear-4D control is capacity-matched, and the penalty of `LSTM+PI-QNN` is a hand-set heuristic rather than a physical constraint; the analyses of the three hybrids other than `LSTM+QNN` were added after the main analysis, and no equivalence margin was specified in advance. An earlier execution of the core experiment with a preprocessing error (two features misaligned across stations) gave RF − `LSTM+QNN` = +0.029 instead of +0.020; the change is smaller than its standard error but shows how much a seed-averaged hybrid comparison can move with a preprocessing detail (Section S1.4).

Label validity. Incomplete months are 23 % of the test partition (Section 4.5). The label scenarios re-evaluate fixed predictors: the models were not retrained under alternative label definitions, months with incomplete coverage among the 12 antecedent months of a window were not examined, and 897 station-months with at least 10 observed precipitation days that do not survive the inner join were not examined.

Baseline tuning and scope. Hyperparameters were set manually and no reproducible tuning history is available (Section S3); an Optuna-tuned Random Forest on 2,273 rows (0.6798 ± 0.0045) scores slightly below the untuned one on the same rows, so tuning moves the AUC by less than the noise, but XGBoost (0.662) should not be read as a ceiling for boosted trees. The panel is small (3,363 station-months), monthly aggregation does not resolve convective episodes, and the chronological split falls in a period of accelerated warming, so with one chronological split we cannot determine how distribution shift modifies model performance or sensitivity to optimization randomness. We evaluate one ansatz family in noiseless simulation with a 0.9–3.9 % quantum share of trainable parameters; inference-time gate noise changes the results little (Section S7.13) but says nothing about noise during training (Wang et al., 2021).

Generality. All empirical conclusions concern one complete-variable panel of four stations. The sampling argument and the resolution rule transfer to other benchmarks (Section 5.4), but the findings about particular models and references do not replace validation on further datasets, preferably ones in which learned models clearly improve on a base-rate reference (Section 5.3).

6 Conclusions

On a benchmark of 528 correlated test station-months, the test sample, not the training seed, limits what can be concluded about the classical models: one AUC has a standard error of about 0.024, the minimum detectable difference of 15-seed means is 0.031–0.044 AUC, and it falls only as $N^{-1/2}$ with the size of the test set. For variational hybrids the seed variance is also substantial, so both sources must be reported. Near this limit the verdict depends on the procedure: DeLong and the year-block bootstrap disagree on Random Forest against the LSTM and against `LSTM+QNN`, and known-truth simulations show that each can be miscalibrated under plausible dependence between test years.

In the quantum case study, four hybrid models fitted on the same rows with 15 seeds lie 0.020–0.026 AUC below Random Forest and are not distinguished from a parameter-matched, circuit-free classical control, the only comparison that isolates the circuit. Resolving the Random Forest–hybrid gap reliably would need about 2.5 times the present test set under DeLong, and resolving whether the circuit adds anything over its control five to seven times. No model, classical or quantum, improves on a month×station climatology,

and within stations the climatology discriminates best. Alternative labels for incomplete months preserve the signs of the differences but not their significance.

The consequence for QML benchmarking is practical: state the estimand, compute the minimum detectable difference before ranking architectures, report seed and sampling error separately, compare against a base-rate reference and a classical control at the same interface, and treat differences below the resolution of the design as unresolved rather than as evidence for either model.

Data and Code Availability

Code, the processed monthly panel and model checkpoints are available at https://github.com/enriquerb32/limits-of-resolution-qml. The daily records are AEMET open data obtained through *datosclima.es*; only the aggregated monthly panel is redistributed (`results/panel_checkpoint.csv`), with attribution to AEMET. The core experiment is produced by `code/local_experiments/exp17_hybrids_multiseed.py` (options `--fix-et0 --p60 28.1`), which stores the test probabilities of every model and seed (`results_hybrids_trainonly_fixet0_p60_28.1_probs.npz`); the classical rerun is archived in `results_rerun_train_only_canonical_probs.npz`. Most other analyses re-evaluate these archived probabilities without retraining, with one script per table in `code/local_experiments/`; the repository README maps every table and figure to its command and output file, and `scripts/build_traceability.py` checks the numerical claims of the paper against the archived results. Internal variable names and the feature glossary are in the Supplement.

Declaration of Generative AI in Scientific Writing

During the preparation of this manuscript, the author used generative AI tooling for editorial language polishing, LaTeX formatting and structural drafting, and for drafting analysis scripts that the author ran and checked. No reported test statistic, empirical metric or benchmark result was generated by AI models; the numerical results come from the released scripts. The author assumes full scientific and editorial responsibility for the content.

Author Contributions

E.R. conceived the study, curated the data, developed the models, performed the statistical analysis, and wrote the manuscript.

Competing Interests

The author declares no competing financial or non-financial interests.

Funding

This research received no external funding.

Acknowledgements

The author acknowledges the State Meteorological Agency of Spain (AEMET) for providing the underlying meteorological station records under open-data terms.

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A Computations

Estimators. For two models scored on the same 528 rows, the DeLong estimator gives the covariance of the two AUCs from their structural components (DeLong et al., 1988); it treats station-months as independent. The block bootstrap resamples the 19 calendar years (2008–2026) with replacement, keeping all stations of a year together, recomputes both AUCs on each resample ($B = 2{,}000$) and takes the standard deviation of the paired difference as its standard error (Künsch, 1989). With 19 clusters the estimate of the standard error is imprecise, and the direction of its discrepancy with the DeLong value depends also on the dependence structure (here the block standard error is below the DeLong value for RF − LSTM: 0.010 versus 0.014 for single seeds, 0.006 versus 0.011 for seed means). The two-way bootstrap draws a calendar-year resample and, independently for each model, a resample of its 15 seeds, and recomputes the difference of seed-averaged AUCs.

Seed-mean statistic. The AUC of one model is the mean of the DeLong structural components over positives and over negatives, so the mean of the 15 AUCs equals the AUC statistic built from components averaged over seeds, and the paired DeLong variance of $\hat{\Delta}$ follows from the averaged components. This treats the 15 seeds as fixed and so conditions on the fitted models; the two-way bootstrap relaxes this by resampling seeds.

Seed share of variance. For model m with seed standard deviation s_m and sampling standard error SE_m, the share is $s_m^2/(s_m^2 + \mathrm{SE}_m^2)$. Using the mean DeLong SE over the 15 seeds, Random Forest (SD 0.0035, SE 0.0232), XGBoost (0.0054, 0.0236) and the LSTM (0.0063, 0.0237) give 2.3 %, 4.9 % and 6.6 % (2 %, 5 % and 7 % in Table 3). For the hybrids of the core experiment, `LSTM+QNN` (SD 0.0098, SE 0.0237), `LbL-QNN` (0.0129, 0.0238), `LSTM+QTL` (0.0147, 0.0239) and `LSTM+PI-QNN` (0.0170, 0.0238) give 14.7 %, 22.7 %, 27.4 % and 33.9 %.

Simulation mechanism. For each replication, years $u = 1, \ldots, 19$ receive a prevalence effect $\tau_\pi\, \eta_u$ on the logit of $\pi = 0.41$ and a number of rows of 528 split as equally as possible (or, in the unequal scenario, a multinomial split with Dirichlet(3) shares). Scorers $k \in \{A, B\}$ are $s_k = \mu_k y + e_k + \tau_s\, w_u + \tau_m\, v_{k,u}$, with $e_k = \sqrt{\rho}\, z + \sqrt{1-\rho}\, \varepsilon_k$ (so that the scorer noise has correlation ρ), w_u a year shift shared by both scorers and $v_{k,u}$ a year shift specific to each scorer; all $\eta_u, w_u, v_{k,u}, z, \varepsilon_k$ are standard normal and $\mu_k = \sqrt{2}\,\Phi^{-1}(\mathrm{AUC}_k)$ without year effects. The design AUCs without year effects are 0.65 for scorer B and $0.65 + \Delta$ for scorer A; the scorer noise has correlation $\rho = 0.9$, and the scorer-specific year shifts $v_{k,u}$ are independent between scorers. Scenario B has $\tau_\pi = \tau_s = 0.3$ and $\tau_m = 0$; C adds $\tau_m = 0.3$; D is C with Dirichlet year sizes; E has $\tau_\pi = \tau_s = \tau_m = 0.6$. The simulation target is the pooled population difference under the mechanism, with years drawn independently and weighted equally, that is, the difference for new years; in scenario D the shares of the 528 rows among years follow a Dirichlet(3) draw that is independent of the year effects, the sample statistic pools observations (so it weights years by their sizes), and the reference is the pooled population AUC with equally likely years of equal size. Scenario D changes the distribution of cluster sizes. Because the sizes are independent of the year effects, the pooled statistic is expected to be centered on the equal-weight reference; this was checked empirically and not proved: in D the mean of the estimates differs from the reference by +0.0002, −0.0008 and −0.0004 for design differences of 0, 0.015 and 0.03 (Monte Carlo standard error of the mean about 0.0008). The population AUC of each scorer is computed as the mean of four independent draws of 400,000 rows from the same mechanism with years drawn independently and equally weighted, and the Monte Carlo standard error of the population difference is reported. When the design difference is 0 the scorers are exchangeable and the population difference is set to 0. DeLong tests use the paired estimator of the seed-free case; the block bootstrap resamples years ($B = 300$ per replication).